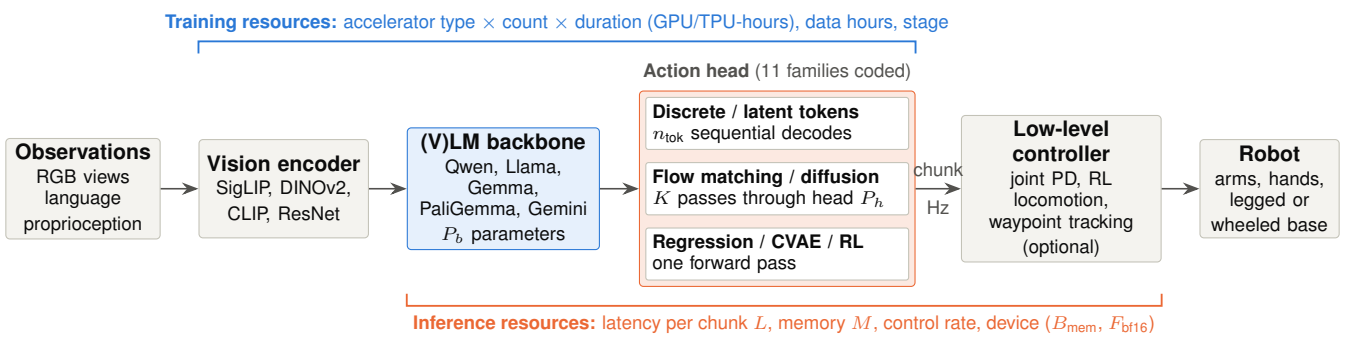
Training resources: accelerator type × count × duration (GPU/TPU-hours), data hours, stage
Observations
RGB views
language
proprioception
Vision encoder
SigLIP, DINOv2,
CLIP, ResNet
(V)LM backbone
Qwen, Llama,
Gemma,
PaliGemma, Gemini
P_b parameters
Action head (11 families coded)
Discrete / latent tokens
n_{tok} sequential decodes
Flow matching / diffusion
K passes through head P_h
Regression / CVAE / RL
one forward pass
chunk
Hz
Low-level controller
joint PD, RL
locomotion,
waypoint tracking
(optional)
Robot
arms, hands,
legged or
wheeled base
Inference resources: latency per chunk L, memory M, control rate, device (B_{mem}, F_{bf16})